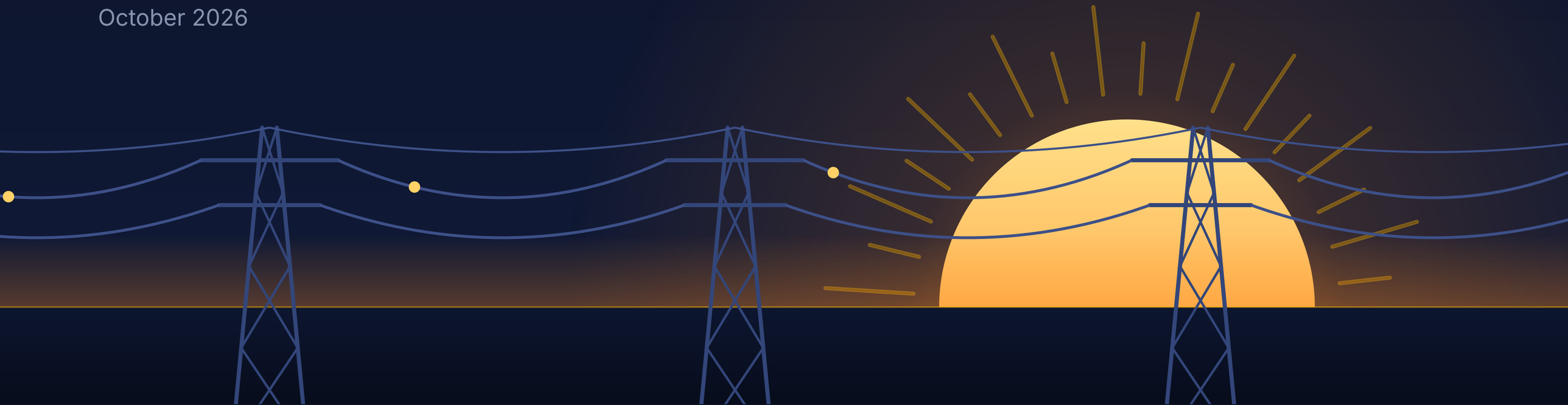
BUS 9430 · CAPSTONE 1 · ISSUE SELECTION AND PLAN

Tomorrow’s Sun, Today’s Plan

Forecasting California's solar power one day ahead

Team: Carolina Lenes Arreola, Juan David Jaimes, Mohammad Mahmud, Vince Lanzillo and Raúl J. Solá Navarro

October 2026



THE BUSINESS PROBLEM

Plan locked today. The sun wasn't consulted.

ILLUSTRATIVE

~80%

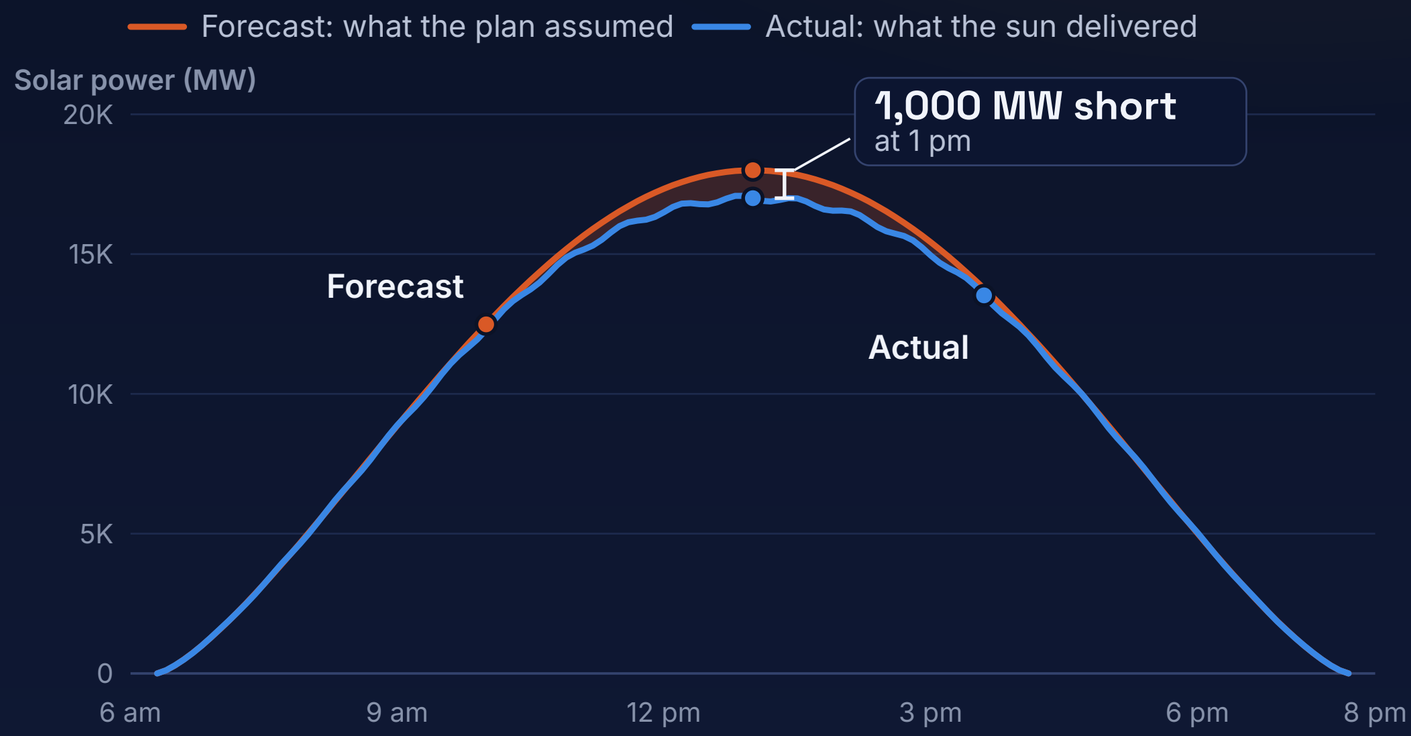
of California's power is run by CAISO (32 million people).

1 day

ahead: the plan is locked using a weather forecast.

~1,000 MW

average miss, even for the pros.



FORECAST TOO HIGH

Backup power bought at the last minute

FORECAST TOO LOW

Gas plants running for nothing

Sources: CAISO Key Statistics, May 2026; Enverus analysis of CAISO's day-ahead solar forecast (error of 1,082 MW in May and 930 MW in June 2026). The chart is an illustration, not real data.

WHAT WE ARE TESTING

REGRESSION: PREDICTING A NUMBER

Can free data get us close to the pros?

INPUT
Weather forecast from the day before

→

OUR MODEL
Predicts tomorrow's solar, hour by hour

→

OUTPUT
Megawatts for every hour of tomorrow

Same days, same scorecard

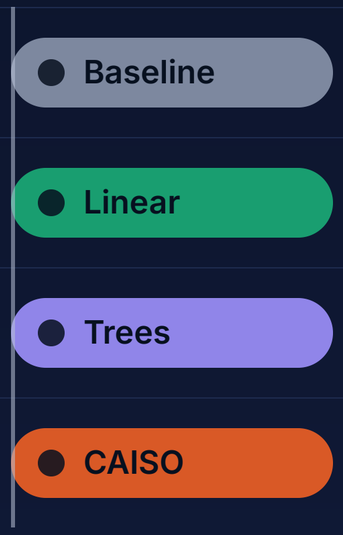
Results: Dec 14

Same as yesterday
The simple guess

Simple model
Easy to explain

Flexible model
Catches complex patterns

CAISO's own forecast
The professional benchmark





Judged on what actually happened, on days our models have never seen.

Data, all public: weather by Open-Meteo.com (CC BY 4.0), solar output from the U.S. EIA, benchmark from CAISO.

WHO IT HELPS

Better forecasts, four winners



Grid operators

A clearer view of when forecasts miss.



Utilities and ratepayers

Less spent on emergency power.



Solar and battery owners

Sharper bids and schedules.



Communities and the climate

Less gas burned as insurance.

HOW WE GET IT DONE CRISP-DM

Six steps, one proven method

1 BUSINESS UNDERSTANDING
Define the problem
Capstone 1

2 DATA UNDERSTANDING
Get the data
Oct 14 to 23

3 DATA PREPARATION
Clean and combine it
Oct 23 to Nov 16

4 MODELING
Build the forecasts
Nov 17 to 27

5 EVALUATION
Test against CAISO
Nov 30 to Dec 8

6 DEPLOYMENT
Share the results
Dec 8 to 14

Learn along the way, loop back, improve

CRISP-DM is the standard six-step process data teams use. Dates are from our project plan.

SUMMARY

Matters.

Cleaner, cheaper grid operations for 32 million people.

Measurable.

A clear benchmark to compare against, every hour, on public data.

Doable.

Free data, nine weeks, three checkpoints.

NEXT STEPS

Oct 14 to 23
Pull and check the data

Nov 16
Capstone 2: data ready

Dec 14
Capstone 3: results

Questions?